

Thermal Control Design Windows for Microring WDM Links in Co-Packaged AI Optical I/O

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Abstract—Silicon microring WDM links in co-packaged AI optical I/O need active resonance control. We derive BER-level detuning windows for PAM4 receivers using thermo-optic drift, $Q=5200$ rings, Lorentzian leakage, and package stress cases. For 100 GHz WDM at 25 dB SNR, lock accuracy is 101–165 pm.

Index Terms—co-packaged optics, silicon photonics, microring resonator, WDM, thermal drift, resonance locking, PAM4

I. INTRODUCTION

Co-packaged photonics is moving optical I/O into the thermal neighborhood of compute packages, where packaging and thermal control become system-level constraints rather than device afterthoughts [1]–[3]. Silicon microring resonators (MRRs) are attractive for dense WDM because they are compact and wavelength-selective, but their resonances shift strongly with temperature. Near 1550 nm, silicon’s thermo-optic coefficient gives $d\lambda/dT \approx 72 \text{ pm}/^\circ\text{C}$ [4], comparable to a substantial fraction of practical MRR linewidths.

Prior work has demonstrated MRR tuning and stabilization [5], and recent WDM rings report measured $Q=5200$ with 589 pm/V gate tunability [6]. What is still missing for CPO link budgeting is a BER-level answer to: how accurately must the resonance be held after thermal drift and control? This paper contributes a residual detuning window, a Q /spacing/SNR design map, and a package-informed translation from multi-nm drift to sub-linewidth lock accuracy.

II. MODEL

Each MRR drop port is modeled as

$$T_{\text{drop}}(\Delta\lambda) = \frac{1}{1 + (2\Delta\lambda/\text{FWHM})^2}, \quad (1)$$

where $\text{FWHM} = \lambda_0/Q$. Temperature drift is

$$\Delta\lambda_T = \frac{\lambda_0}{n_g} \frac{dn}{dT} \Delta T, \quad (2)$$

with $\lambda_0 = 1550 \text{ nm}$, $n_g = 4.0$, and $dn/dT = 1.86 \times 10^{-4} \text{ K}^{-1}$ [4]; thermal expansion is neglected. Eight equal-power WDM channels are placed on 50, 100, or 200 GHz grids; 100 GHz is 0.80 nm near 1550 nm. For a center drop port, all seven non-selected WDM interferers are Lorentzian-weighted, and we report the worse of positive/negative residual detuning.

We use nonnegative Gray-coded IM/DD PAM4 levels $\{0, 1, 2, 3\}$. SNR is $\text{Var}(x)/\sigma^2$ at the slicer for an aligned

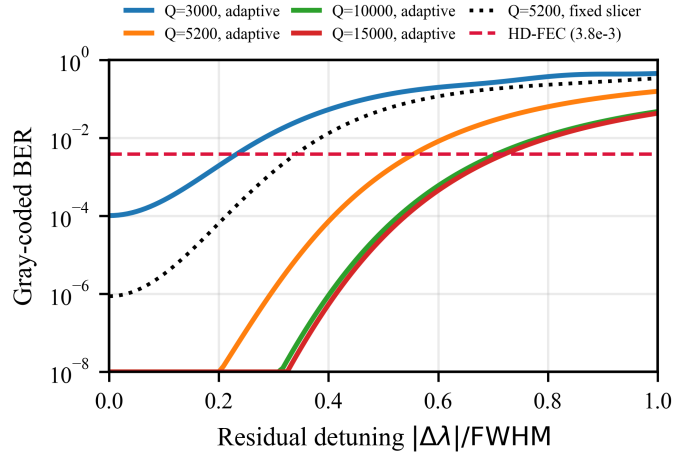


Fig. 1. Gray-coded BER versus residual resonance detuning for 100 GHz WDM and 25 dB electrical slicer SNR under identical Lorentzian-only receiver assumptions. Solid curves use adaptive mean-pedestal tracking; the dotted curve shows the conservative $Q=5200$ fixed-threshold receiver.

unit-gain channel. The adaptive case tracks the slowly varying mean WDM-leakage pedestal; the fixed-threshold case bounds receivers without pedestal adaptation. BER is computed by enumerating discrete WDM interference and integrating AWGN over PAM4 decision regions. Monte Carlo counts true bit errors. Package stress uses imported CSV data or a disclosed synthetic edge-hotspot scenario motivated by OIF and Ansys CPO workflows [7], [8]. The design criterion is the nominal pre-FEC HD-FEC $\text{BER} = 3.8 \times 10^{-3}$.

TABLE I
MODEL SCOPE AND HEADLINE VALUES.

Item	Included/value	Scope/role
MRR/WDM	Lorentzian, 7 interferers	crosstalk
PAM4 BER	Gray bits, AWGN	slicer SNR
Not modeled	OMA/RLM, TIA/RIN, EQ	link penalties
$Q=5200$ FWHM	298 pm	linewidth
Lock budget	101–165 pm	fixed/adaptive

III. RESULTS

Fig. 1 shows that the useful control variable is not raw temperature but residual detuning after compensation. At 100 GHz spacing and 25 dB slicer SNR, the literature-reported $Q=5200$ case remains below HD-FEC to 0.55 FWHM, equal

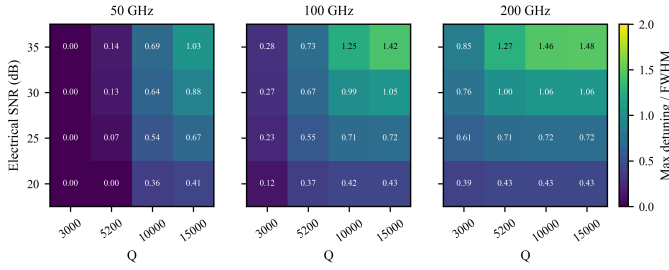


Fig. 2. Residual detuning, in linewidths, that preserves the nominal pre-FEC HD-FEC threshold for 50/100/200 GHz WDM.

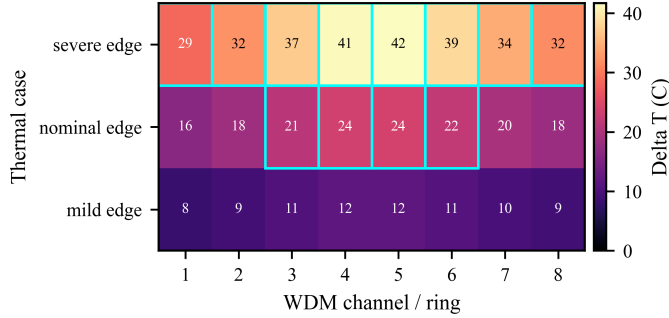


Fig. 3. Illustrative package-informed thermal stress cases for an eight-ring optical engine. Cell values are local photonic-layer ΔT in $^{\circ}\text{C}$; cyan outlines mark $\Delta T \geq 20.8^{\circ}\text{C}$, corresponding to $> 1.5 \text{ nm}$ drift at $72 \text{ pm}/^{\circ}\text{C}$.

to 165 pm or 2.3°C equivalent drift. Without mean-pedestal tracking, the same case tightens to 0.34 FWHM, or 101 pm. Lower- Q rings are more tolerant in absolute linewidth but suffer more WDM leakage; $Q=3000$ allows only 0.23 FWHM at the same SNR.

The spacing/SNR/ Q design window in Fig. 2 quantifies this tradeoff. At 50 GHz, low- Q rings have no FEC-compliant residual window under these assumptions because Lorentzian tails dominate even at zero detuning. At 100 GHz, the $Q=5200$ window grows from 0.37 to 0.73 FWHM over 20–35 dB SNR. At 200 GHz, crosstalk is relaxed and the lock budget becomes mainly noise- and linewidth-limited.

Fig. 3 illustrates how the BER window maps onto package-scale temperature nonuniformity. OIF estimates that a 3.2 Tb/s dense optical engine can dissipate 32 W in a $20 \times 20 \text{ mm}^2$ footprint [7], while Ansys demonstrates thermal-map-driven WDM BER simulation [8]. Our disclosed synthetic edge-hotspot cases span 8–42 $^{\circ}\text{C}$ local rise; the 1.5 nm ITiO range [6] is exceeded in the nominal center channels. These illustrative cases show why coarse multi-nm trim and 101–165 pm closed-loop residual control should be budgeted separately.

Finally, Fig. 4 cross-checks the BER implementation against Monte Carlo symbol streams under the same Lorentzian, equal-power, and AWGN assumptions. Agreement near the HD-FEC crossing confirms that the window is not a symbol-error artifact. The model remains first-order: SNR does not separately model OMA, RLM, photodiode responsivity, TIA noise/bandwidth, RIN, equalization, or control-loop dynamics.

IV. CONCLUSION

Control-aware BER modeling shows that uncompensated silicon MRR WDM links are not a credible CPO operating

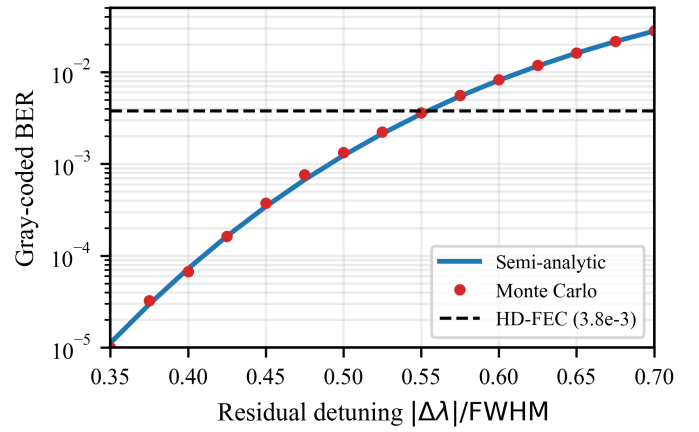


Fig. 4. Near-FEC Monte Carlo cross-check for $Q=5200$, 100 GHz WDM, and 25 dB slicer SNR. Markers count Gray-coded bit errors over 200,000 PAM4 symbols per point.

mode, but active control can be budgeted quantitatively. For the literature-reported $Q=5200$ WDM ring at 100 GHz spacing and 25 dB slicer SNR, the residual lock budget is about 100–165 pm depending on pedestal tracking. Synthetic package-informed stress cases illustrate why practical CPO optical I/O needs coarse tuning range and closed-loop stability.

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